

Generation Recall, Not Verification, Binds LLM Structure Elucidation from Literature Spectra

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Abstract

Frontier large language models are often presented as near-solved structure elucidators on curated spectra. We ask a harder, more operational question: given the molecular formula and literature-reported IR / ^1H / ^{13}C peak lists — not digitised traces — can an off-the-shelf LLM recover the correct constitution, and which stage fails when it does not?

We introduce IRSpectra-Bench, a blind, mechanically scored peak-list benchmark of 194 compounds drawn from redistributable experimental band lists (IRexp; companion Data Descriptor) (Yabbarov et al., 2026), with a fixed RDKit InChIKey-connectivity scoring contract and decomposable generation recall / verification precision metrics. On IRSpectra-Bench, a frontier LLM recovers the correct constitution for 28% top-1 (95% CI 22–35), or 15% once reweighted to corpus composition. The bottleneck is candidate proposal, not verification: the true structure enters the pool for only 34% of compounds, and where it does, training-free forward-verification selects it 89% of the time (58/65). The same recall $\ll$ precision asymmetry replicates across four vendor families. A formula-only control drops top-1 from 23% $\rightarrow$ 5%; accuracy is flat in source-paper year. Generating wider lifts recall 32% $\rightarrow$ 42% and top-1 23% $\rightarrow$ 30% on 60 compounds, while verification alone moves whole-benchmark top-1 only 28% $\rightarrow$ 30%. Recovered from published top- k figures, recall carries 68–83% of the accuracy collapse when systems move from curated or simulated data to real heterogeneous spectra. Frozen predictions, scorers and code are released for mechanical re-evaluation.

1 Introduction

Structure elucidation from spectra sits at the centre of synthetic and analytical chemistry, and machine learning increasingly treats it as a sequence or agent task. Encoder-decoder models train on paired corpora (Chacko et al., 2024; Ottomano et al., 2025; Alberts et al., 2024); LLM agents and puzzle benches report high recovery on curated or education spectra (Kamber, 2026; Guo et al., 2024; Zhuang et al., 2025; Espejo Morales et al., 2026). Cross-paper numbers swing wildly with harness (e.g. GPT-4o from 1.4% to 57.8% on MolPuzzle under different prompting (Guo et al., 2024; Zhuang et al., 2025)). What remains under-measured is the chemist’s operational setting: take peak lists as printed in open-access papers and recover constitution, with a fixed molecular-representation scoring contract and an honest account of which stage of elucidation binds.

We argue that end-to-end top-1 alone is the wrong reporting primitive. Any system that returns a ranked candidate list factorises as

$$\text{top-1} = \underbrace{P(\text{true} \in \text{pool})}_{\text{generation recall}} \times \underbrace{P(\text{rank} = 1 \mid \text{true} \in \text{pool})}_{\text{verification precision}}. \quad (1)$$

These rates have different denominators and must not be differenced. Prior work already notes that re-ranking cannot recover a missing candidate (Priessner et al., 2026); we supply the missing measurement infrastructure at scale.



Figure 1: Diagnosis on IRSpectra-Bench ($n=194$): generation recall, not verification, is the bottleneck. Where the true structure is never proposed, no ranker can recover it; where it is proposed, verification usually selects it.

Contributions.

1. IRSpectra-Bench — a blind, complexity-stratified peak-list benchmark ($n=194$) with a pre-registered InChIKey-connectivity scorer and community reporting contract (top-1, generation recall, verification precision | recall).
2. A recall-bound diagnosis on a frontier LLM: 34% proposed; 89% selected once proposed; top-1 moves only 28% $\rightarrow$ 30% under training-free forward-verification.
3. Contamination and cross-vendor controls — formula-only and recency bounds; four-vendor replication of recall $\ll$ precision.
4. A literature decomposition that recovers the same split from published top- k figures, showing recall carries most of the collapse on real heterogeneous data.

Dataset pointer (not this paper’s primary object). Experimental band lists come from IRExp (121,233 records; 43,060 structure-linked; 33,201 full IR+ $^1\text{H}+^{13}\text{C}$ +structure quadruples), released on Hugging Face (ilkhamfy/IRExp). Construction, licensing, and technical validation of the corpus are the subject of a companion Scientific Data manuscript (in preparation) (Yabbarov et al., 2026). This ICLR paper cites that resource and does not re-present a Data Descriptor.

2 Related work

Trained spectra $\rightarrow$ structure models. Sequence and graph decoders on paired corpora (Chacko et al., 2024; Hu et al., 2024; Yang et al., 2026; Ottomano et al., 2025; Alberts et al., 2024; 2025) are accurate in-distribution but typically scored end-to-end. We do not claim to beat them on IRSpectra-Bench; Spectro, NMIRacle, Alberts IR transformers and CASE are not yet scored on our bench (Section 7) — a gap we state honestly and leave to the released scorer.

LLM agents and puzzle benches. ChemCrow (M. Bran et al., 2024), Coscientist (Boiko et al., 2023), SpectralLM (Su et al., 2026), SpecMol (Shen et al., 2025), tree-search elucidation (Zhuang et al., 2025), IR-Agent (Noh et al., 2025) and Priessner et al. (Priessner et al., 2026) improve how a model reads spectra or re-ranks a fixed pool. We fix one protocol and ask which stage binds once the spectra have been read.

Contamination-aware and heterogeneous-data benches. Most prior suites use simulated or single-instrument spectra. IRSpectra-Bench uses literature-reported peak lists across thousands of laboratories and reports formula-only and recency controls (Section 5.2). Espejo Morales et al. (Espejo Morales et al., 2026) compare education vs industrial instrument files (80.9% $\rightarrow$ 20.6%); they ask which architecture, we ask which stage binds on published reports.

Forward verification and CASE. Matching predicted to observed shifts — DP4 (Smith & Goodman, 2010), CASE (Elyashberg et al., 2012), NMR-Solver (Jin et al., 2026), NMRAgent (Fang et al., 2026a) — is easier than inverse generation. CASE enumerators have

exhaustive recall by construction; the LLM literature almost never reports whether the true structure was proposed at all. We decompose error rather than chase a single top-1.

Positioning. Concurrent 2026 suites (MolQuest (Han et al., 2026), SpecX (Xiang et al., 2026), NMRGym (Fang et al., 2026b)) bound related tasks. Ours is an openly redistributable suite on literature peak lists with a stage decomposition measured on Claude and replicated across three other model families.

3 IRSpectra-Bench

3.1 Task and inputs

Each of 194 problems supplies the molecular formula (as from HRMS), the IR band list (cm^{-1}), and ^1H / ^{13}C shift lists — exactly the textual form reported in open-access papers. No name, SMILES or scaffold hint. Inputs are author-transcribed band and shift lists, not absorbance traces or FIDs: a different object from digitised-spectrum benchmarks (Espejo Morales et al., 2026), and one that matches how language models consume literature.

Problems are drawn from the structure-complete split of IRExp (irexp_resolved; see companion Data Descriptor / HF card) (Yabbarov et al., 2026). In 10/194, author peak assignments name a ring system; those are solved 1/10 vs 54/184 for the rest, so annotation does not carry the headline.

3.2 Difficulty, balance, and scoring contract

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridgehead system, ≤ 22 heavy atoms; else complex (98/96). The released benchmark is 50/50 by design; the eligible corpus is 17.5% simple / 82.5% complex. Reweighted to corpus composition, top-1 falls 28.4% $\rightarrow$ 15.2% and recall 33.5% $\rightarrow$ 19.8%. We report both; the reweighted figure is the better estimate for an arbitrary paper, the 50/50 figure the comparable leaderboard number.

Scoring. A prediction is correct if RDKit InChIKey connectivity (first 14 characters) matches — we score constitution. Full stereo-sensitive InChIKey gives 21.1% top-1 vs 28.4% constitution. External submissions: up to three ranked SMILES per qid, no structure hints (docs/LEADERBOARD.md; scripts/score_submission.py).

Primary metrics (report all three).

metric	definition
Top-1	best-ranked candidate matches at connectivity layer
Generation recall	reference in the candidate pool before re-ranking
Verification precision recall	correct top-1 among recall-positive compounds

CIs are bootstrap 95%; paired model differences use McNemar with Holm correction.

4 Experimental setup

Solver. Frontier LLM agent (Claude Opus via consumer subscription), closed-book, one sub-agent per batch, up to three ranked candidates, RDKit formula check only. No fine-tuning, no paid API for the core protocol. Inference is not exactly reproducible (no pinned snapshot, temperature or seed — Section 7); scoring is: frozen predictions, ground truth and scorers regenerate every training-free number.

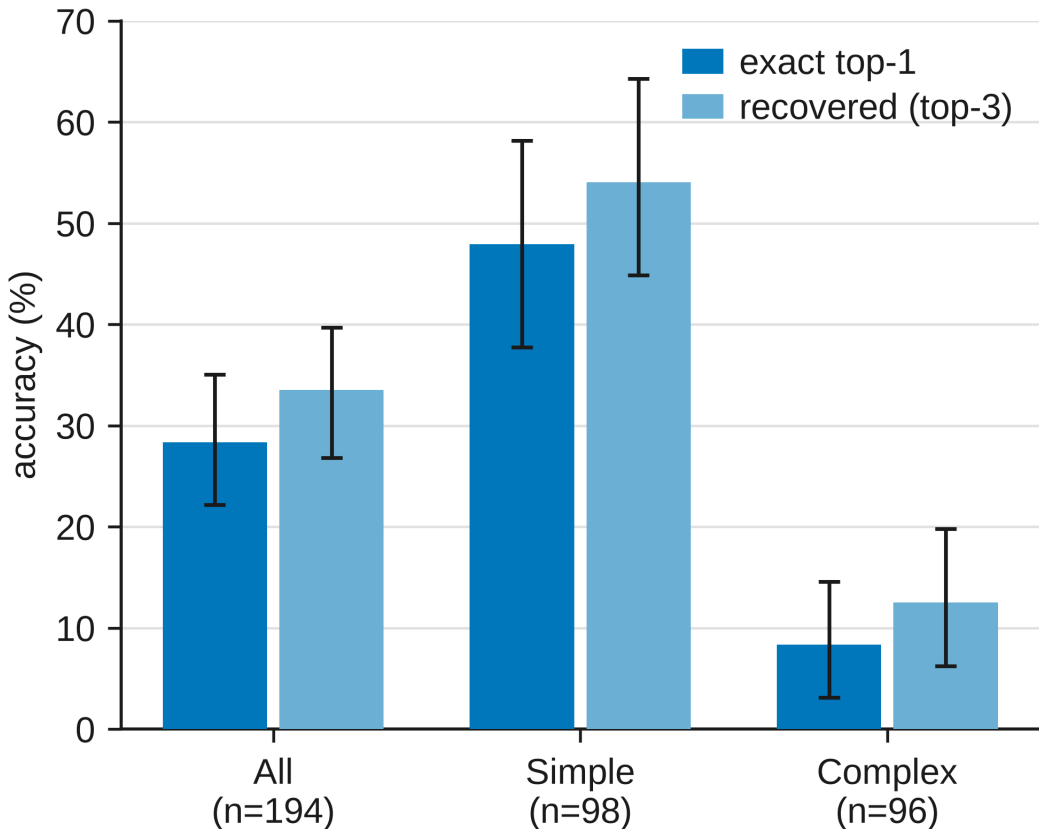


Figure 2: Top-1 and recovered accuracy by difficulty ($n=194$), with bootstrap 95% CIs.

Table 1: Headline elucidation on IRSpectra-Bench ($n=194$).

metric	overall ($n=194$)	simple ($n=98$)	complex ($n=96$)
top-1 exact constitution	28.4% [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto ≥ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

Forward-verification loop (Section 5.4). Generate candidates $\rightarrow$ forward-predict each candidate’s ^{13}C (blind to observed spectrum) $\rightarrow$ re-rank by symmetric chamfer distance between predicted and observed ^{13}C peak sets.

Controls and probes. Formula-only / recency (Section 5.2); four-vendor replication on a 60-compound arm (Section 5.3); generate-wide sampling; HOSE lookup and GNN ^{13}C verifiers; small IRexp-fine-tuned generator probe (fenced from headline claims).

5 Results

5.1 Headline performance

Accuracy falls with size: 60.5% top-1 at ≤ 15 heavy atoms, 28.3% at 16–25, 7.0% above 25. Of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; only 22.6% share the true Murcko scaffold — wrong connectivity at the right composition, not mostly fine regiochemistry.

Table 2: Formula-only control (paired, $n=60$).

condition	top-1	recovered (top-3)
formula only	3/60 (5%)	3/60 (5%)
formula + IR + ^1H + ^{13}C	14/60 (23%)	19/60 (32%)

Table 3: Cross-vendor decomposition ($n=60$). Recall and precision have different denominators — the claim is the inequality, not a difference.

model	generation recall	verification precision recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

A 24-compound four-model Claude comparison is capability-sensitive (strict nesting Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable) but underpowered for adjacent ranks and carries a disclosed protocol asymmetry (Section 7). A battery-electrolyte subset ($n=46$) matches the headline regime (26% top-1; recall-bound).

5.2 Is the model reading the spectra?

Outcomes are perfectly nested (McNemar $p=0.001$). Accuracy is flat in source-paper year across all 194 ($r=-0.007$), bounding pretraining recall of specific papers. We claim a strong bound, not exclusion of contamination (Section 7).

5.3 Cross-vendor replication

Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol solved the same 60-compound arm under the identical protocol. Verification precision exceeds generation recall in every arm; bootstrapping separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional.

Candidate budgets differ (Claude mean 2.20 vs others 3.00), so recall rankings are approximate. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$).

5.4 Forward-verification decomposition

When the true structure is among the candidates, forward-verification selects it in 58/65 cases (89%). Where a choice existed ($n=37$), it gets 30/37 (81%) against a 54.0% derangement floor (+27.1 points). The margin over self-ranking is small and unresolved (seven gained, four lost; McNemar $p=0.55$) — we do not claim an accuracy advance. Precision is high while recall binds: no re-ranking repairs the 129/194 compounds where the true structure was never proposed (Figure 1).

Generate-wide. Ten solver agents, up to six candidates each, pooled and re-ranked on the 60-compound arm: recall 32% $\rightarrow$ 42%, top-1 23% $\rightarrow$ 30% (McNemar $p=0.34$); verification precision falls 84% $\rightarrow$ 72%. The training-free ceiling remains recall-limited.

Non-LLM verifiers. On the same candidate sets, a HOSE-style lookup ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%) — directional. Deranging observed ^{13}C pairings collapses precision ($p=0.001$). A small IReXP-fine-tuned generator pooled with Claude lifts recall 33.5% $\rightarrow$ 54.1% and top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$), showing the wall is a property of training-free elicitation, not of the task.

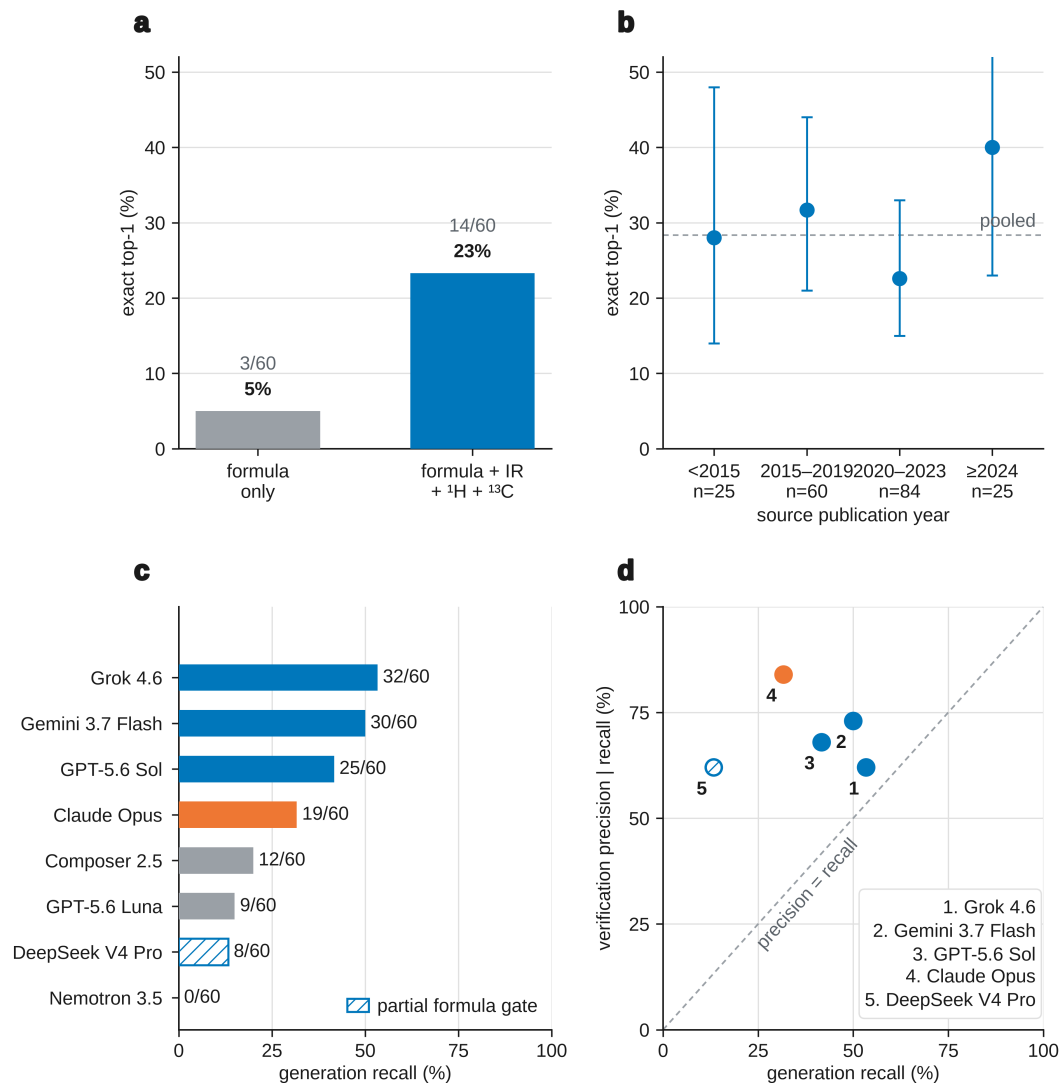


Figure 3: Robustness of the recall-bound diagnosis: formula-only vs full modality; accuracy vs year; cross-vendor recall and precision.

Table 4: Forward-verification decomposition.

	60-compound arm	full benchmark ($n=194$)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
precision recall — forward-verification	16/19 (84%)	58/65 (89%)
multi-candidate only — forward-verification	10/13 (77%)	30/37 (81%)

5.5 Decomposition across published literature

Any system reporting $\text{top-1} = a$ and $\text{top-}k = b$ implies $\text{recall} \geq b$ and conditional precision $\leq a/b$. Applying this to published figures (scripts/literature_decomposition.py):

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse.

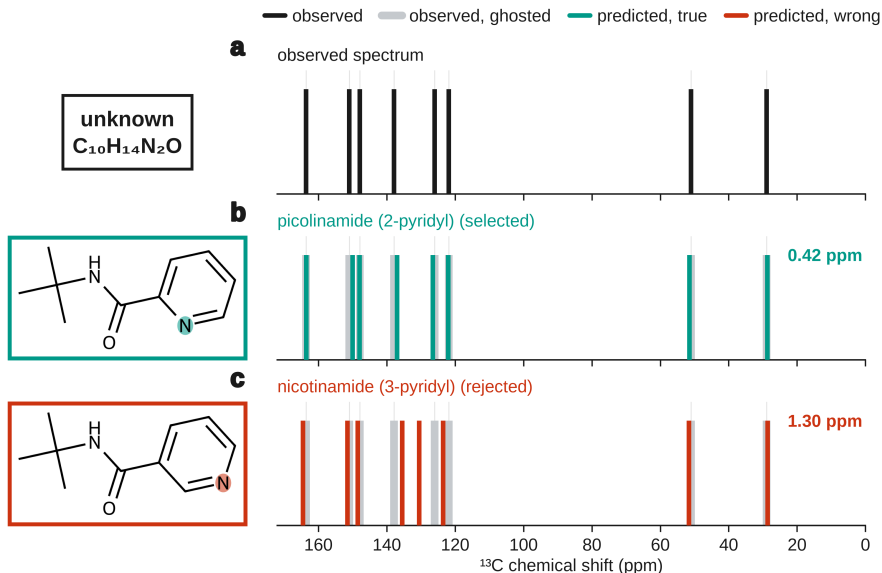


Figure 4: Forward-verification on a regioisomer pair: inverse task is ambiguous; forward-predicted ^{13}C separates isomers.

Table 5: Selected literature decomposition rows (connectivity-scored or as published).

system	data (n)	top-1	top-k (k)	recall	prec.
this work, solver alone	literature (194)	28.4%	33.5% (3)	33.5%	84.6%
this work + forward-verify	literature (194)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver et al., 2026)	(Jin literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent et al., 2026a)	(Fang literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales (Espejo Morales et al., 2026)	education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales (Espejo Morales et al., 2026)	industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, atoms et al., 2025)	6-13 NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, split (Xiang et al., 2026)	scaffold simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

Recall binds where spectra are real and heterogeneous; ranking dominates on simulated or single-library data. The field almost never reports whether the true structure was proposed at all.

6 Discussion

Frontier LLMs are good verifiers ($\approx 89\%$ conditional precision) and weak proposers ($\approx 34\%$ recall) on real literature peak lists. The primary research contribution of this paper is the benchmark + stage decomposition + contamination/cross-vendor diagnosis, not a solved elucidator and not a Data Descriptor for IRExp.

Reporting contract. External work on IRExp-Bench should deposit ranked SMILES under the released protocol and report, at minimum: (i) top-1 constitution, (ii) generation recall, (iii) verification precision | recall, with bootstrap CIs and candidate budget. Top-

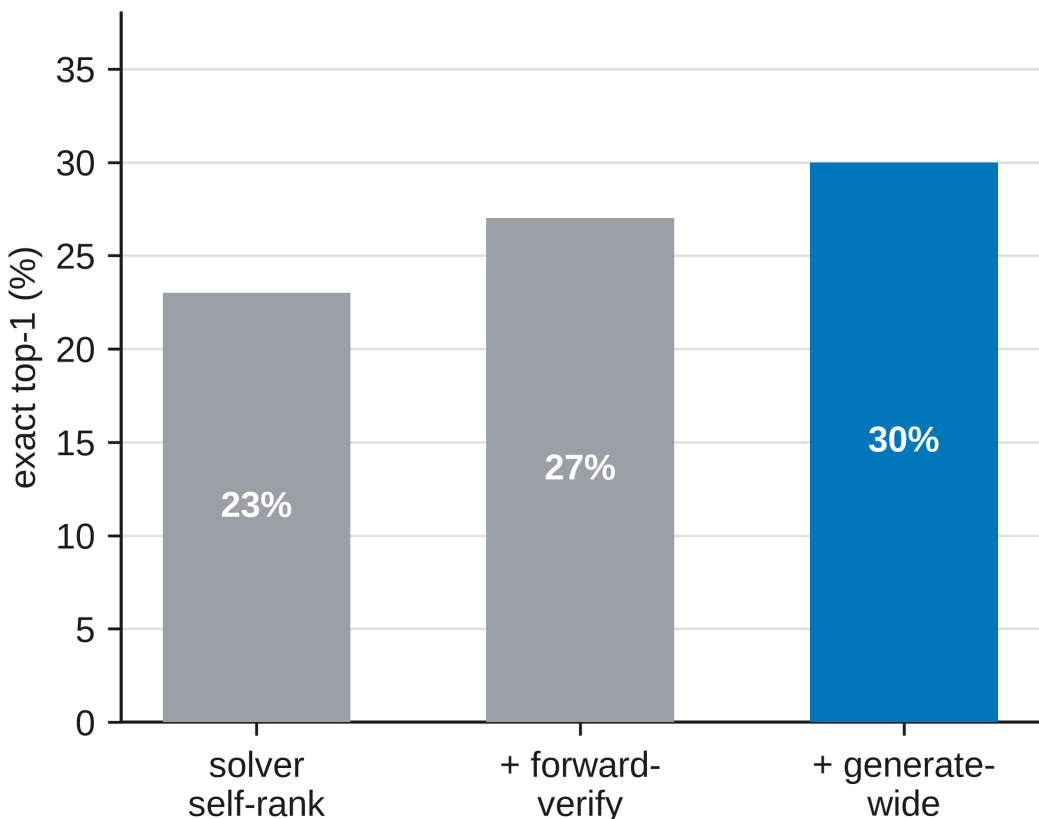


Figure 5: Forward-verification inference ladder on the 60-compound arm.

1 alone conflates the two stages. Nonsignificant ladder gains (forward-verification vs self-ranking, $p=0.55$; generate-wide top-1, $p=0.34$) should be cited as diagnostic, not as accuracy advances. Corpus-reweighted top-1 (15.2%) is the better estimate for an arbitrary literature draw.

Training is not foreclosed: IRexp fine-tuning lifts recall to 54%. What compounds is open experimental data, honest benchmarks, and inference scaffolding that rides each new model.

7 Limitations

Scoring is mechanical; solver runs were transcript-audited for zero ground-truth access. (i) Consumer harness. No model snapshot, temperature, seed or thinking tier; exact inference is not reproducible. Scoring is. Instruction text wrapping Claude batches was not captured — only per-compound payloads are released. (ii) Pretraining contamination is bounded by formula-only (23% $\rightarrow$ 5%) and flat recency but not excluded; verbatim spectral strings from PMC in the prompt remain a retrieval confound. (iii) Object type, formula and scoring. Band/shift lists, not raw traces; formula supplied (as from HRMS) — absolute accuracies are not interchangeable with no-formula systems. Headline metrics score constitution; with stereochemistry, top-1 is 21.1%. (iv) Statistical honesty. Forward-verification vs self-ranking unresolved ($p=0.55$); generate-wide top-1 likewise ($p=0.34$). Four-model comparison at $n=24$ underpowered for adjacent ranks with disclosed protocol asymmetry. The inference ladder diagnoses a bottleneck; it is not a demonstrated accuracy advance. (v) Missing on-bench cheminformatics baselines. Spectro, NMIRacle, Alberts IR transformers and CASE were not scored on IRSpectra-Bench — the LLM recall wall is not cleanly separated from harness or modality choice. (vi) Cross-vendor scope. Headline $n=194$ is Claude

Opus; four-vendor replication is on 60 compounds; candidate budgets differ. (vii) Deferred arms. Expert-chemist audit of elucidation outputs formally deferred (not run). Leave-one-modality-out ablation abandoned for this manuscript. Extraction-recall human audit of the mining parser is deferred to the Scientific Data companion. (viii) Scope. Battery subset uses literature electrolyte chemistry, not operando spectra; single-sample scoring per compound; organic literature bias of PMC-OA sources.

8 Conclusion

IRSpectra-Bench makes LLM structure elucidation from literature peak lists measurable and decomposable. On that protocol, generation recall — not verification — binds end-to-end accuracy; the diagnosis replicates across vendors and recovers from published top- k figures. We release frozen predictions and a mechanical scorer so others can report the same three numbers. The redistributable experimental band-list corpus behind the bench is documented separately as a Scientific Data Data Descriptor (in prep.) (Yabbarov et al., 2026) and mirrored at [ilkhamfy/IRexp](https://github.com/ilkhamfy/IRexp).

Reproducibility statement

Every training-free number regenerates from released questions, answers, frozen predictions and scripts in the project repository. The sampler, InChIKey scorer and forward-verification harness are scripted end-to-end. Consumer-harness inference cannot be bit-exact replayed (Section 7); distributional agreement is the intended claim. Fine-tuning against the bench should use `data/train_no_bench.jsonl.gz` so benchmark InChIKeys are withheld.

Ethics statement

This work evaluates publicly available open-access literature peak lists and commercial LLM APIs under consumer terms. No human-subject data. We do not claim clinical or regulatory readiness for automated elucidation. Licensing of redistributed numeric extractions is documented in the companion Data Descriptor and dataset notices (PMC-OA terms are mixed — not uniformly CC-BY; Chemotion is CC-BY-SA).

Acknowledgements

Deferred to camera-ready.

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A Dataset pointer (IReXp)

IReXp is not re-described here as a Data Descriptor. For counts, provenance, licensing caveats, transcription audit (560/560 bands on $n=60$), and file schema, see:

- Hugging Face: <https://huggingface.co/datasets/ilkhamfy/IReXp>
- Companion manuscript: Scientific Data Data Descriptor (in preparation) (Yabbarov et al., 2026)
- Combined archive (pre-split JCIM-shaped paper): docs/archive/combined_PAPER.md
- Feasibility audit: docs/irexp_scientific_data_audit.md

Benchmark construction details beyond Section 3 (spectral validation filters, prompt-leakage audit, electrolyte SMARTS) live in docs/BENCHMARK.md and the Electronic Supplementary Information of the combined archive.

B Extended tables and figures

Supplementary figures currently shared with the combined manuscript under docs/figures/: fig0_overview, fig2_size, fig4_dataset, fig5_models, fig6_electrolyte, fig7_crossvendor, fig_contamination, fig_generator_probe, fig_verifier. Full artefact inventory: repository README / docs/archive/combined_PAPER.md Data availability.